🔴 3. Multi-Operator D2D Communication

✔️ What is Multi-Operator D2D?

- Communication between devices belonging to **different network operators**

👉 Example:
📱 Airtel ↔ 📱 Jio

- Multi-operator D2D allows **devices of different network operators to communicate directly** without routing through core network.
- Direct communication between devices
- Works across **different operators**
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✔️ Why Needed?

- Users from different networks need direct communication
- Important in:
  - Emergency situations
  - Public safety networks

✔️ Requirements

- Standard communication protocols
- Secure authentication
- Operator cooperation

✔️ Challenges

1. Security Issues

- Risk of data leakage

2. Privacy Problems

- User data must be protected

3. Billing Issues

- Who charges for data?

✔ Solutions

- Encryption techniques
- Secure login/authentication

✔ Advantages

- Enables universal communication
- Improves connectivity
- Useful in emergency networks

✔ Example

- During disaster, all users communicate regardless of network

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✔ Diagram (Exam)

Operator A User ↔ Operator B User

Feature	NOMA	OFDMA	Multi-Operator D2D

Basic Concept	Power domain multiplexing	Frequency domain multiplexing	Direct device communication
Communication Type	Base station to users	Base station to users	Device to device
Access Type	Non-orthogonal	Orthogonal	Direct communication
Domain Used	Power domain	Frequency domain	Proximity domain
Spectral Efficiency	Very high	High	Moderate to high
Latency	Low	Moderate	Very low
Scalability	Very high	High	Moderate
Coverage	Depends on BS	Depends on BS	Limited (short range)
Power Allocation	Required	Not required	Not required